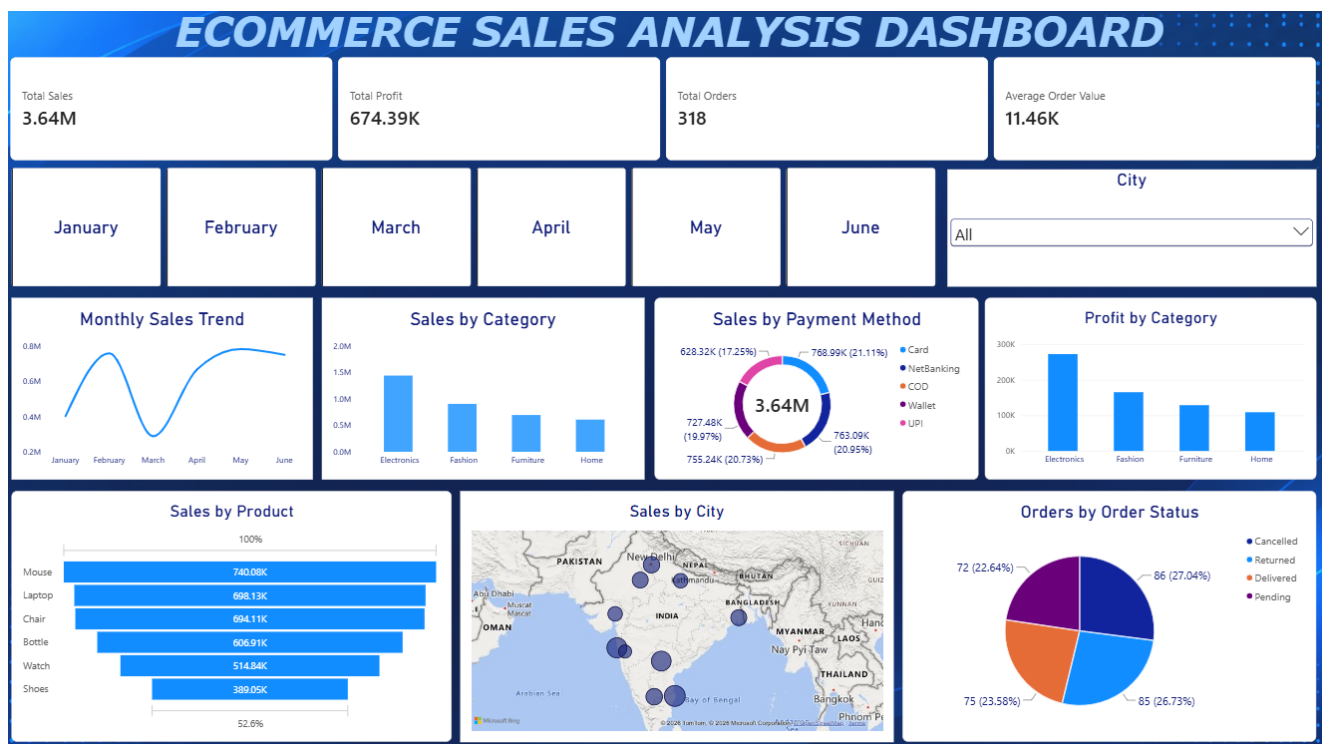


E-COMMERCE SALES ANALYSIS DASHBOARD

PROJECT REPORT

Interactive Power BI dashboard for analyzing sales, profit, orders, product performance, payment methods, city performance and order status.

TOTAL SALES 3.64M	TOTAL PROFIT 674.39K	TOTAL ORDERS 318	AVG. ORDER VALUE 11.46K
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Final Power BI dashboard

1. Project Overview

This project presents an interactive e-commerce sales dashboard built in Power BI. The dashboard consolidates key commercial metrics into a single view and allows users to explore performance by month and city. The design focuses on quick business interpretation, clean KPI tracking and consistent sales calculations.

2. Project Objectives

- Track overall sales, profit, order volume and average order value.
- Identify sales trends across months and compare performance by product and category.
- Understand how customers pay and how order statuses are distributed.
- Visualize geographic sales activity by city.
- Create an interactive dashboard with month and city filtering for faster analysis.

3. Dataset Structure

The e-commerce orders dataset contains transaction-level fields used for sales, customer, product, date, location and profitability analysis. Key fields visible in the model include:

Area	Fields
Transaction / customer	Transaction_ID, Buyer_Name, Email, Phone
Product	Item_Name, Category, Quantity, Selling_Price
Revenue & profit	Sales, Net_Amount, Discount, Profit
Date	Purchase_Date, Delivered_Date, Month, Week_Day, Year
Location	City, State
Order details	Order_Status, Payment_Method

4. Data Modeling and DAX Logic

The dashboard uses explicit measures for core KPIs and a helper month number to ensure chronological sorting. The final sales definition was aligned across the KPI and sales visuals so that totals reconcile consistently.

Total Sales	SUM('ecommerce orders'[Sales])
Average Order Value	DIVIDE([Total Sales], [Total Orders])
MonthNo	MONTH('ecommerce orders'[Purchase_Date])

5. Dashboard Design and Visuals

The report follows a three-level information hierarchy: KPIs at the top, trend/category analysis in the middle and detailed product/geographic/order-status analysis at the bottom. A blue dashboard theme with white visual cards keeps the focus on the data.

Visual	Purpose
KPI Cards	Total Sales, Total Profit, Total Orders and Average Order Value provide a fast executive summary.
Month Slicer	January to June is sorted chronologically using MonthNo.
City Dropdown	Allows the full dashboard to be filtered to a selected city.
Monthly Sales Trend	Shows month-to-month sales movement and makes peaks and dips easy to identify.
Sales by Category	Compares revenue contribution across Electronics, Fashion, Furniture and Home.
Sales by Payment Method	Donut chart compares Card, NetBanking, COD, Wallet and UPI.
Profit by Category	Compares profitability across product categories.
Sales by Product	Ranks key products using a horizontal funnel-style bar visual.
Sales by City	Map visual shows city-level activity with visible bubbles.
Orders by Order Status	Shows the share of Cancelled, Returned, Delivered and Pending orders.

6. Key Findings

- Total sales reached 3.64M, with total profit of 674.39K across 318 orders.
- Average order value is 11.46K after aligning the KPI with the same Total Sales definition used throughout the dashboard.
- Electronics is the strongest category in both sales and profit, making it the most important category in the current data.
- The payment mix is relatively balanced: Card leads at 768.99K (21.11%), followed closely by NetBanking at 763.09K (20.95%) and COD at 755.24K (20.73%).
- Among the displayed products, Mouse leads with 740.08K in sales, followed by Laptop at 698.13K and Chair at 694.11K.
- Order-status distribution is also balanced: Cancelled 86 (27.04%), Returned 85 (26.73%), Delivered 75 (23.58%) and Pending 72 (22.64%).
- The monthly trend shows noticeable volatility, with February and May among the stronger months and March showing a clear dip.

7. Business Interpretation

The dashboard suggests that growth opportunities should focus on maintaining strength in Electronics while investigating the causes of weaker months and the high combined share of cancelled and returned orders. Payment behavior is diversified rather than concentrated in one method, so checkout improvements should support all major payment options rather than prioritizing only one.

8. Recommended Business Actions

- Investigate cancellation and return reasons because these two statuses together represent more than half of the displayed orders.
- Protect inventory and promotional support for top-selling products such as Mouse, Laptop and Chair.
- Analyze the March sales dip by city, category and product to identify the underlying driver.
- Track profit alongside sales when planning promotions so high-revenue categories also contribute healthy margins.
- Use the city filter to compare regional demand and identify locations suitable for targeted campaigns.

9. Project Skills Demonstrated

Power BI Dashboard development	DAX KPI measures	Data Modeling Sorting and field design	Data Visualization Business-focused charts
Data Validation Metric reconciliation	Interactivity Slicers and filters	Business Analysis Insight generation	UI Design Consistent layout/theme

10. Suggested GitHub Repository Structure

- Ecommerce_Sales_Analysis_Dashboard.pbix - Power BI dashboard
- Ecommerce_Sales_Analysis_Project_Report.pdf - project documentation
- README.md - project summary, screenshots and instructions
- data/ - cleaned or sample dataset (if sharing is permitted)
- screenshots/ - final dashboard image

11. Conclusion

The final dashboard provides a compact, interactive view of e-commerce performance and demonstrates the full reporting workflow from metric definition and sorting logic to dashboard design and business interpretation. It is suitable as a portfolio project because the KPIs reconcile, the visuals answer distinct business questions and the report supports clear decision-making.